ISSN: 0256-307X

中国物理快报
Chinese Physics Letters

Volume 26 Number 4 April 2009

A Series Journal of the Chinese Physical Society
Distributed by IOP Publishing

Online: http://www.iop.org/journals/cpl
http://cpl.iphy.ac.cn

JUST FOR AUTHORS
— CHINESE PHYSICS LETTERS

CHINESE PHYSICAL SOCIETY

CHIN. PHYS. LETT. Vol. 26, No. 4 (2009) 043401

Electron-Helium Scattering in a Bichromatic Laser Field in the Second-Order Born Approximation *

ZHOU Bin(周斌)[1**], LI Shu-Min(李书民)[1,2***]

[1] *Department of Modern Physics, University of Science and Technology of China, PO Box 4, Hefei 230026*
[2] *Institut für Theoretische Physik, Universität Heidelberg, Philosophenweg 16, 69120 Heidelberg, Germany*

(Received 15 December 2008)

The elastic scattering of electrons in atomic helium assisted by a bichromatic laser field is investigated in the second order Born approximation. The target atom is approximated by a simple screening potential. The dependence of the differential cross section on the relative phase between the two laser components is calculated, and compared with the recent results of first order Born approximation [Sun J F, Liang M C and Zhu Z L 2007 Chin. Phys. Lett. ***24*** *2572].*

PACS: 34.80.Qb, 32.80.Wr, 34.50.Rk, 34.80.Bm

The scattering of electrons by an atomic system in the presence of a laser field has been investigated by many authors since the pioneering work of Bunkin and Fedorov.[1] Summaries of these investigations can be found in the book by Mittleman,[2] the survey by Joachain,[3] and the review by Ehlotzky.[4] As the experimental technology improved, powerful lasers and new kinds of laser fields have been applied to laser-assisted atomic and molecular processes. Due to the significance in dynamic control of the multicolor lasers, the processes they modify have received considerable attention. By changing the phase difference between the frequency components of the multicolor laser, we can enhance or modify the atomic and molecular processes, which are called coherent phase control. This possibility stimulated a considerable body of research work.[5−15] In Ref. [15], Sun *et al.* considered the intense bicromatic laser assisted elastic scattering of electrons in helium in the first Born approximation (FBA). As an improvement of the theoretical treatment, in this Letter, we calculate the same process in the second Born approximation (SBA), accompanied by the transfer of a few photons, and compared with the FBA treatment. The atomic units $\hbar = m = e = 1$ are used through out the derivation.

Consider the free-free transition in the helium atom in the presence of a bichromatic laser field. Since in a laser beam the density of radiation quanta is so large that the depletion of this beam by emitting or absorbing quanta from it is negligible, we may describe the laser field classically[2] as a superposition of two frequency components ω and 2ω, i.e., $\mathcal{E}(t) = \mathcal{E}_0[\sin(\omega t) + \sin(2\omega t + \varphi)]$, where $\mathcal{E}_0$ is the electric field amplitude, and φ is the relative phase, which it is supposed can be arbitrarily changed. Nevertheless, the laser frequency ω and intensity I are sufficiently low so that the excitations of atomic transitions can be neglected, and the atomic target can be described by a short range potential[16,18]

$$V(r) = -\frac{Z}{r}\sum_{i=1}^{2} A_i \exp(-\alpha_i r), \tag{1}$$

where r denotes the position of the electron with respect to the target nucleus, and Z is the nuclear charge number. For helium, $A_1 = -0.2259$, $A_2 = 1.2259$, $\alpha_1 = 5.5272$, and $\alpha_2 = 2.3992$. The continuum state of the electron can be described by the Volkov wave function

$$\chi_{\boldsymbol{k}} = \exp(i\boldsymbol{k}\cdot\boldsymbol{r})\exp\Big[-iEt - \frac{i}{\omega^2}\boldsymbol{k}\cdot\boldsymbol{\mathcal{E}}_0\sin(\omega t)\Big] \cdot \exp\Big[-\frac{i}{4\omega^2}\boldsymbol{k}\cdot\boldsymbol{\mathcal{E}}_0\sin(2\omega t + \varphi)\Big], \tag{2}$$

where $\boldsymbol{k}$ is the electron wave vector, and E is the electron energy.

The scattering matrix for the laser-assisted free-free transition in the second Born approximation reads

$$S_{fi}^{\langle 2\rangle} = S_{fi}^{(1)} + S_{fi}^{(2)}, \tag{3}$$

where

$$S_{fi}^{(1)} = -i\int_{-\infty}^{\infty} dt\chi_{\boldsymbol{k}_f}^*(\boldsymbol{r},t)V\chi_{\boldsymbol{k}_i}(\boldsymbol{r},t), \tag{4}$$

$$S_{fi}^{(2)} = -i\int_{-\infty}^{\infty} dt\int d\boldsymbol{r}\int_{-\infty}^{t} dt'\int d\boldsymbol{r}'\chi_{\boldsymbol{k}_f}^*(\boldsymbol{r},t) \cdot V(r)G(\boldsymbol{r},t;\boldsymbol{r}',t')V(r')\chi_{\boldsymbol{k}_i}(\boldsymbol{r}',t), \tag{5}$$

where $G(\boldsymbol{r},t;\boldsymbol{r}',t')$ is the Green function

$$G(\boldsymbol{r},t;\boldsymbol{r}',t') = -\frac{i}{(2\pi)^3}\int d\boldsymbol{k}\chi_{\boldsymbol{k}}(\boldsymbol{r},t) \cdot \chi_{\boldsymbol{k}}^*(\boldsymbol{r}',t')u(t-t'), \tag{6}$$

*Supported by the National Nature Science Foundation of China under Grant Nos 10874169 and 10674125, the National Basic Research Program of China under Grant No 2007CB925200, and the Deutsche Forschungsgemeinschaft.

**Email: binzhou@mail.ustc.edu.cn
***Email: lism@ustc.edu.cn

© 2009 Chinese Physical Society and IOP Publishing Ltd

JUST FOR AUTHORS — CHINESE PHYSICS LETTERS

where $u(x)$ is the step function.

Using the potential of Eq. (1) and wave functions like in Eq. (2), the matrix element $S_{fi}^{\langle 2\rangle}$ is then cast into

$$S_{fi}^{\langle 2\rangle} = -2\pi i \sum_l T_{f,i}^{\langle 2\rangle}(l)\delta(E_f - E_i + l\omega), \tag{7}$$

where $T_{f,i}^{\langle 2\rangle}(l)$ is the transition matrix element, resolved with respect to the multiphoton exchange processes

$$T_{f,i}^{\langle 2\rangle}(l) = T_{f,i}^{(1)}(l) + T_{f,i}^{(2)}(l). \tag{8}$$

Here the first order term reads

$$T_{f,i}^{(1)}(l) = B_l\Big(\lambda, \frac{1}{4}\lambda, \varphi\Big)V(\boldsymbol{k}_f, \boldsymbol{k}_i), \tag{9}$$

where

$$V(\boldsymbol{k}_f, \boldsymbol{k}_i) = \int d\boldsymbol{r} e^{-i(\boldsymbol{k}_f - \boldsymbol{k}_i)\cdot \boldsymbol{r}} V(r), \tag{10}$$

is the Fourier transformation of the potential, and

$$B_l\Big(\lambda, \frac{1}{4}\lambda, \varphi\Big) = \sum_{n=-\infty}^{\infty} J_{l-2n}(\lambda) J_n\Big(\frac{1}{4}\lambda\Big)\exp(-in\varphi), \tag{11}$$

the generalized Bessel function with $\lambda = (\boldsymbol{k}_f - \boldsymbol{k}_i)\cdot \boldsymbol{\mathcal{E}}_0/\omega^2$. $J_n(\lambda)$ is the ordinary Bessel function. The second-order term in Eq. (8) reads

$$\begin{aligned} T_{f,i}^{(2)}(l) = {} & \frac{1}{(2\pi)^3}\sum_m \int d\boldsymbol{k} \frac{1}{E_i - E - m\omega + i0^+} \\ & \cdot V(\boldsymbol{k}_f, \boldsymbol{k}) V(\boldsymbol{k}, \boldsymbol{k}_i) B_m\Big(\lambda_1, \frac{1}{4}\lambda_1, \varphi\Big) \\ & \cdot B_{l-m}\Big(\lambda_2, \frac{1}{4}\lambda_2, \varphi\Big), \end{aligned} \tag{12}$$

where $\lambda_1 = (\boldsymbol{k} - \boldsymbol{k}_i)\cdot \boldsymbol{\mathcal{E}}_0/\omega^2$, $\lambda_2 = (\boldsymbol{k}_f - \boldsymbol{k})\cdot \boldsymbol{\mathcal{E}}_0/\omega^2$.

The different cross section (DCS) for the net exchange of l photons between the colliding system and the bichromatic laser field reads

$$\frac{d\sigma}{d\Omega} = \Big(\frac{1}{2\pi}\Big)^2 \frac{k_f}{k_i}\big|T_{f,i}^{\langle 2\rangle}(l)\big|^2. \tag{13}$$

For the numerical calculation, the angle between the polarization vector $\boldsymbol{\mathcal{E}}_0$ and the momentum $\boldsymbol{k}_i$ of the incident electron is chosen to be $\psi_0 = 38°$ as in Ref. [15]. The momentum $\boldsymbol{k}_f$ of the scattered electron is in the plane defined by the polarization vector $\boldsymbol{\mathcal{E}}_0$ and the momentum $\boldsymbol{k}_i$ of the incident electron. The bichromatic laser parameters are $\omega = 0.117\,\text{eV}$ and $\mathcal{E}_0 = 2.7\times 10^8\,\text{V}\cdot\text{cm}^{-1}$.

Figure 1 shows the φ-dependence of the DCS for the transfer of a photon between the electron-helium system and the bicromatic laser field at a scattering angle $\theta = 13°$. The impact energy of the electron is $E_i = 9.5\,\text{eV}$. The phase angle φ varies from 0° to 360°. We observe that our SBA results are 2∼3 times higher than that of the FBA. The results for emission of a photon ($l = 1$) attain the minimum at $\varphi = 180°$, while the results for absorption of photons ($l = -1$) attain maximum. The obvious discrepancy between the SBA and FBA results may attribute to the fact that in the FBA, one only considers the electron transition from the initial state $\boldsymbol{k}_i$ to the final state $\boldsymbol{k}_f$. While in our second Born approximation, not only the transition to the final state $\boldsymbol{k}_f$ is taken into account, but also the transition to the intermediate states $\boldsymbol{k}$ is included. All the results are symmetric about $\varphi = 180°$.

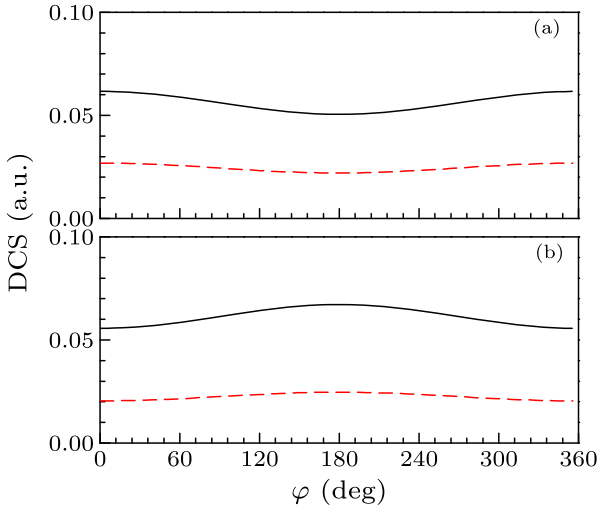

Fig. 1. The relative phase dependence of the DCS in the bichromatic laser assisted electron-helium scattering for (a) the emission of a photon ($l = 1$) and (b) the absorption of photons ($l = -1$). The scattering angle of the electron is $\theta = 13°$, and the impact energy is $E_i = 9.5\,\text{eV}$. The two frequency components of the bichromatic laser field are respectively $\omega = 0.117\,\text{eV}$ and $2\omega = 0.234\,\text{eV}$. The field amplidude is $\mathcal{E}_0 = 2.7\times 10^8\,\text{V}\cdot\text{cm}^{-1}$. Solid curve: results for SBA; dashed curve: results for the FBA (see Ref. [15]).

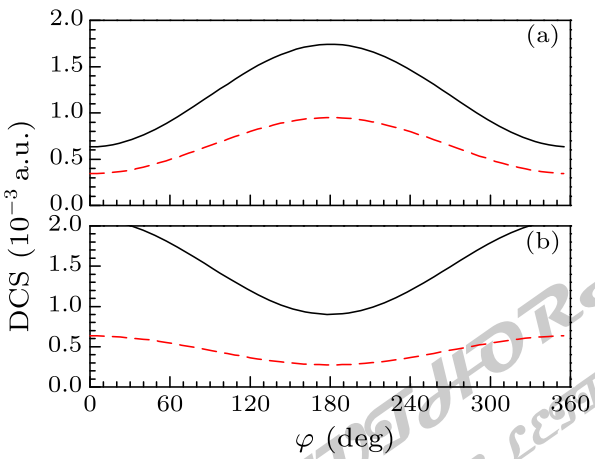

Fig. 2. The φ-dependence of the DCS for (a) the emission ($l = 2$) and (b) the absorption of two photons ($l = -2$) at a scattering angle $\theta = 8°$. The impact energy of the electron and the parameters of lthe aser are the same as in Fig. 1. Solid curve: results for the SBA; dashed curve: results for the FBA.

JUST FOR AUTHORS
— CHINESE PHYSICS LETTERS

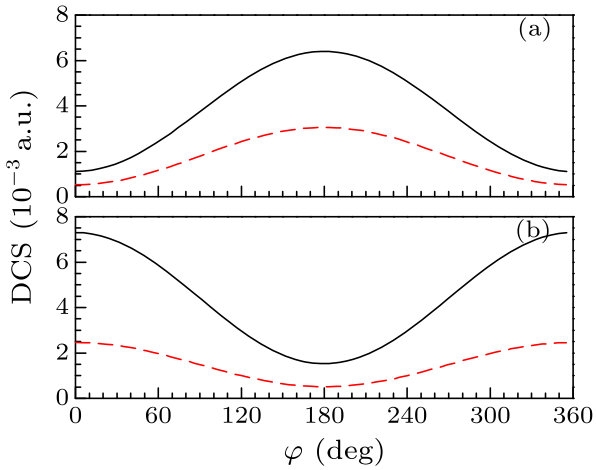

Fig. 3. The same as in Fig. 2, but the scattering angle is $\theta = 13°$.

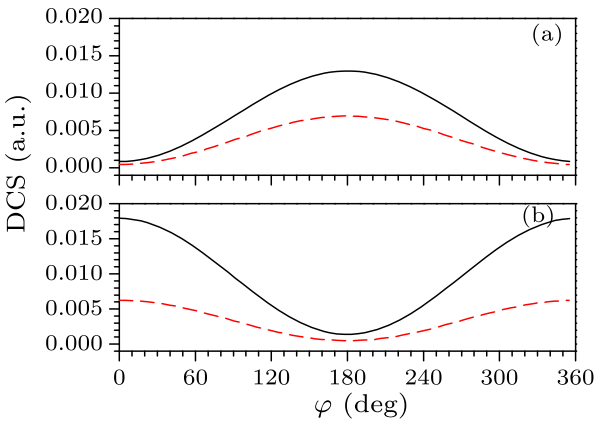

Fig. 4. The same as in Fig. 2, but the scattering angle is $\theta = 13°$, and the impact energy $E_i = 19.5\,\text{eV}$.

In Figs. 2–4, we display the results for the transfer of two photons at some scattering angles and impact energies. In these cases, the DCS is much larger and more sensitive to φ than the FBA results. The variation tendency is opposite to the one-photon case. The DCS for emission ($l = 2$) and absorption of two photons ($l = -2$), respectively, attain the maximum and minimum at $\varphi = 180°$. The large discrepancy with the FBA results suggests that the intermediate states play a more important role in the two-photon processes than in the one-photon case.

In summary, we have investigated the electron-helium elastic scattering in a bichromatic laser field for a simple screening potential in the second Born approximation. The dependence of the DCS on the relative phase is calculated and compared with the FBA results. The significant discrepancy with the FBA results may be attributed to the transition to the intermediate states.

References

[1] Bunkin F V and Fedorov M V 1966 *Sov. Phys. Rep.* **22** 844
[2] Mittleman M H 1982 *Introduction to the Theory of Laser-Atom Interactions* (New York: Plenum)
[3] Joachain C J, Dörr M and Kylstra N 1998 *J. Adv. At. Mol. Opt. Phys.* **42** 225
[4] Ehlotzky F 2001 *Phys. Rep.* **345** 175
[5] Ghalim M and Mastour F 1999 *J. Phys.* B **32** 3783
[6] Zhang S T, Chen J and Li S M 2002 *Can. J. Phys.* **80** 969
[7] Protopapas M, Knight P L and Burnett K 1994 *Phys. Rev.* A **49** 1945
[8] Kamiński J Z and Ehlotzky F 1994 *Phys. Rev.* A **50** 4404
[9] Varró S and Ehlotzky F 1995 *J. Phys.* B **28** 1613
[10] Zhang S T and Chen J 2001 *J. Phys.* B **34** 2201
[11] Varró S and Ehlotzky F 1995 *J. Phys.* B **30** 1061
[12] Varró S and Ehlotzky F 1993 *Phys. Rev.* A **47** 715
[13] Cionga A, Ehlotzky F and Zloh G 2001 *J. Phys.* B **34** 2057
[14] Miloševic D B and Ehlotzky F 1997 *Phys. Rev.* A **56** 3879
[15] Sun J F, Liang M C and Zhu Z L 2007 *Chin. Phys. Lett.* **24** 2572
[16] Salvant F, Martínez J D, Mayol R and Parellada J 1987 *Phys. Rev.* A **36** 467
[17] Sun J F, Zhang S G, Jiang Y H and Yu G Q 1998 *Phys. Rev.* A **58** 2225
[18] Zhu Z L, Jiang Y H and Sun J F 2000 *Chin. Phys.* **9** 419

JUST FOR AUTHORS
— CHINESE PHYSICS LETTERS